

Base 5 Optical Music Encoding System

Value	0	1	2	3	4
Signal	All High	High R	High G	High B	All Low
Color	White	Red	Green	Blue	Black

Paper Strip Layout

1		2 3		4		5 6		7		8	
Instrument 1				Instrument 2				Instrument 3			
Opcode		Immediate		Opcode		Immediate		Opcode		Immediate	
9		10		11 12		13		14		15	
Instrument 4				System				Checksum		Marker	
Opcode		Immediate		Opcode		Immediate					

Instrument Instruction Set

Format (Base 5): [Opcode: 1 digit] [Immediate: 2 digits]

Name	Opcode	Description	Special
Play Note	0 - 2	Play((Opcode * 25) + toDecimal(Immediate) + OFFSET)	000: No-op
Set Instrument	3 - 4	Instrument = ((Opcode - 3) * 25) + toDecimal(Immediate)	

System Instruction Set

Format (Base 5): [Opcode: 1 digit] [Immediate: 2 digits]

Behavior: Latched. State persists until new command overwrites it.

Name	Opcode	Description	Special
Set Tempo*	0 - 1	Tempo = ((Opcode * 25) + toDecimal(Immediate) + 1) * 2	000: No-op
Set Volume**	2 - 3	Target = Immediate Digit 1, Volume = (((Opcode - 2) * 5) + Immediate Digit 2 + 1) * 10	
Set Articulation	4	Target = Immediate Digit 1, State = Immediate Digit 2	

* Sets playback speed to (Tempo)% from 4% to 100% in 2% increments.
** Sets target volume to (Volume)% from 10% to 100% in 10% increments.

Target

State	Name
0	All
1	Instrument 1
2	Instrument 2
3	Instrument 3
4	Instrument 4

Articulation

State	Name	Release Condition
0	Staccato	Release at 1/2 step
1	Normal	Release at 1 step
2	Legato	Release upon non-note instruction (<i>No-op</i> or <i>Set Instrument</i>) or <i>Play Note</i> command with a different pitch
3	Sustain	Release upon <i>Play Note</i> command with a different pitch
4	Infinite	Release when the identical note is triggered again

* *Set Articulation* command immediately releases all active notes on the targeted channel(s) before the new (or same) state is applied.

Checksum Implementation

Format (Base 5): [Checksum: 1 digit]

Calculation	Checksum = (5 - (sum(data_0_to_14) mod 5)) mod 5
Validation	sum(data_0_to_15) mod 5 == 0